SECTION 1 · CONDUIT HUB · PERSISTENT AGENT

1 SYSTEM OVERVIEW

1.1 INTRODUCTION.

jbot is the Conduit durable agent (pi-durable) running as one always-on agent per hub. Underneath, it is a harness thread in a loop with tools, like any chat. What makes it an assistant is the set of parts around that loop, which decide when it runs, what it knows and what reaches the person (figure 1-1).

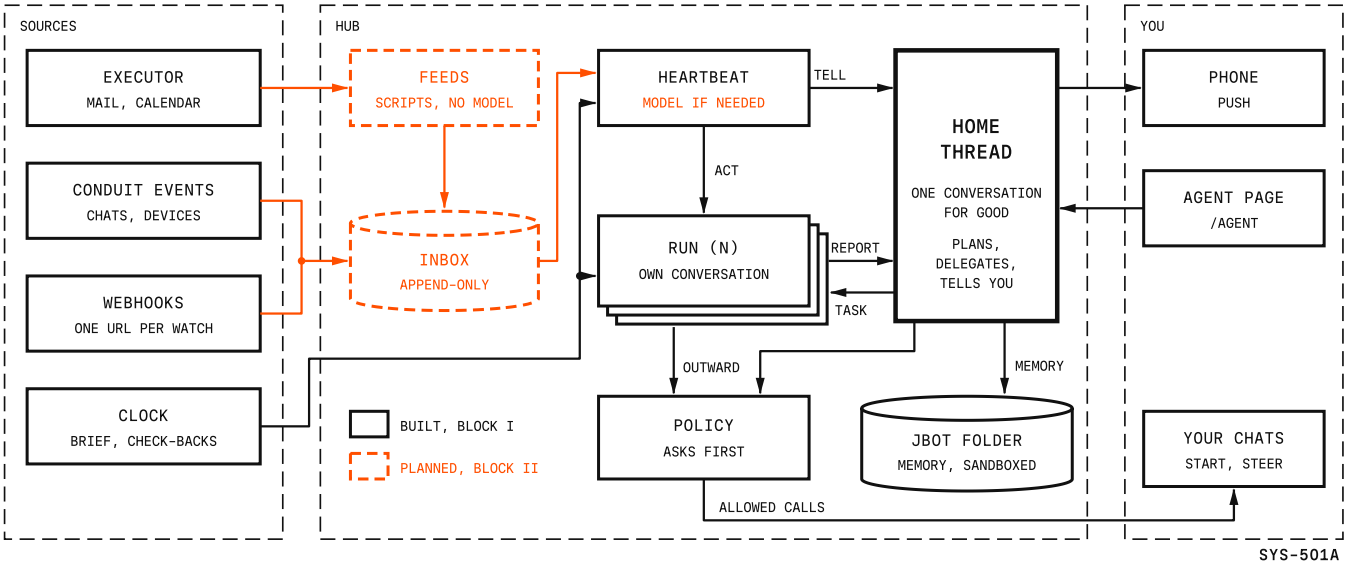


Fig. 1-1 Hub Functional Block Diagram

1.2 FUNCTIONAL DESCRIPTION.

Sources reach the hub as feeds, Conduit events, webhooks and the clock. Feeds append entries to the inbox; the heartbeat reads them and stays quiet, starts runs or tells jbot. The home thread talks with the person and delegates; runs do the work in their own conversations and report back. Every call that leaves jbot's folder passes the policy, which asks the person first (figure 1-2).

1.2.1 Two Blocks.

Block I is built (27 Oct 2026): home thread, runs, schedules, watches, memory files, approvals and push. Block II is the plan of 29 Oct 2026: feeds, the inbox, a heartbeat that calls a model only when there is something to decide, and night upkeep. Block II parts are drawn dashed, in orange.

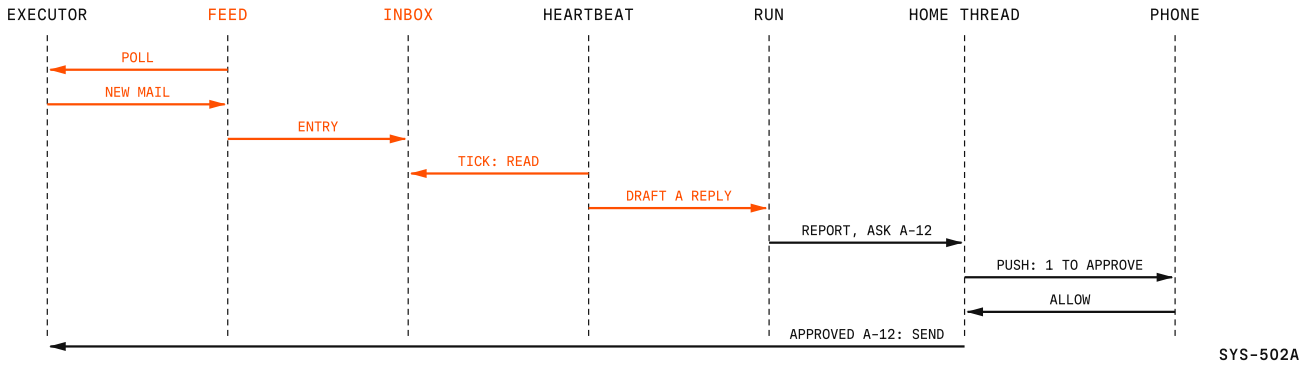


Fig. 1-2 Path of One Email, From Arrival to Send

Table 1-1 System Defaults

PARAMETER	VALUE
Agents per hub	1
Morning brief	08:00 daily
Heartbeat, Block I	every 120 min
Heartbeat tick, Block II	30 min, model if needed
Check-back interval	15 to 10,080 min
Daily budget	US$2.00 (0 = no limit)
Speaks up (dial)	Quiet, Normal, Chatty
Approval lifetime	24 h

NOTE

jbot remembers what the person tells it. It never digests the person's other chats on its own; it reads one only when asked to.

RUN A TASK	YOU
1 SIDEBAR - Chat, jbot	
2 RUN A TASK - select	
3 TASK - type the brief	
4 RUN - press (ctrl+enter)	
√ Working on - shows the run	
5 HOME THREAD - read report	

SYS

SECTION 2 · WORK DONE AWAY FROM THE CONVERSATION

2 RUNS, ROUTINES, HEARTBEAT

2.1 INTRODUCTION.

Anything that is not conversation is a run: a subagent conversation with a brief, a model and effort, a tool scope and a budget, which must end by saying how much its result matters. Routines are saved triggers that start runs; the brief and the heartbeat are routines (figure 2-1).

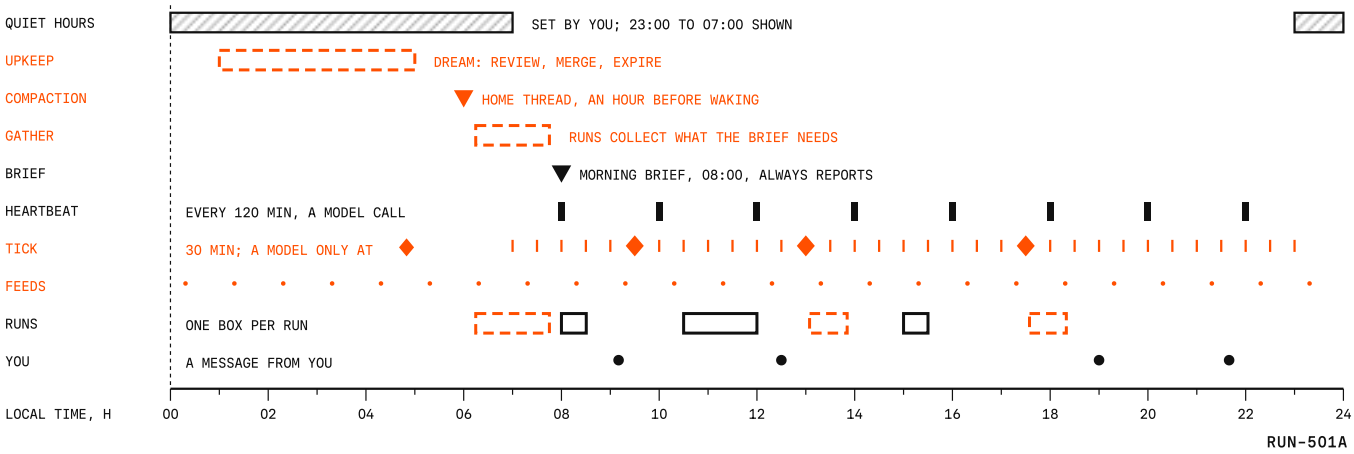


Fig. 2-1 One Day: Upkeep at Night, Brief at Waking, Heartbeat While Awake (Example)

2.2 FUNCTIONAL DESCRIPTION.

A trigger (the clock, a Conduit event, a webhook, a check-back jbot set for itself, or the person through RUN A TASK) spawns a run as a child conversation, on the model and effort set for its kind. The run works alone; an outward call it makes becomes an ask held for the home thread. Its answer's first word says how much the result matters (figure 2-2).

2.2.1 Where Results Go.

Block I posts a report the dial lets through as one line in the home thread, gives jbot a turn on it and pushes once. Block II appends every result to the inbox instead: an urgent entry wakes jbot at once, the rest wait for the heartbeat, and a quiet run leaves only its history.

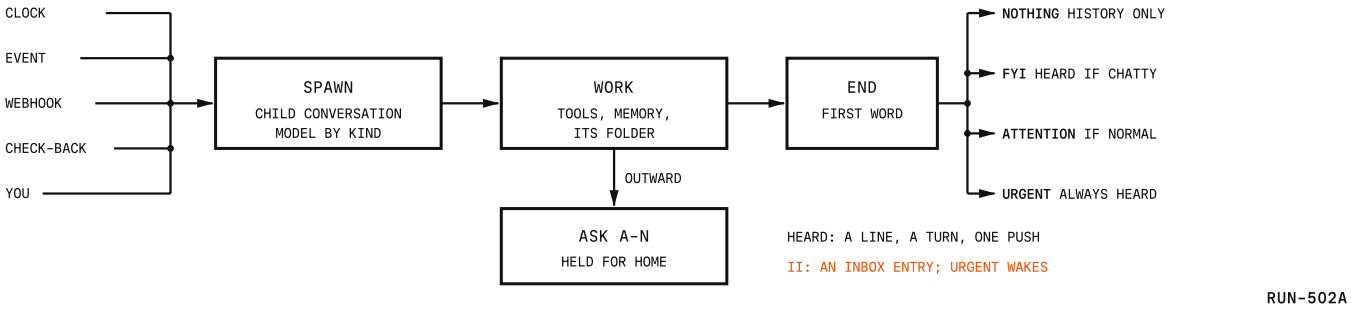


Fig. 2-2 Life of a Run, Trigger to Ending

Table 2-1 What a Routine Records

FIELD	HOLDS
Trigger	daily at, every N min, once; event; webhook
Brief	the task the run is given
Model, effort	per kind: brief, heartbeat, other
Tool scope	the tools the run may use
Ending	report, quiet or failed, by a tool call
Made by	you; or jbot, approved by you
History	each run: transcript, cost, report

CAUTION

The heartbeat never runs in the home thread. Checks that share the chat's conversation let unverified content into long-term memory.

HEARTBEAT TICK HUB

1 INBOX - read unhandled
2 LOOPS - read the due ones
√ none - stop, no model
3 MODEL - decide
4 ACT - runs, check-backs
5 TELL - jbot, if it matters
6 ENTRIES - mark handled

RUN

SECTION 3 · WHAT IT KNOWS AND WHAT IT SEES

3 MEMORY AND CONTEXT

3.1 INTRODUCTION.

jbot's memory is a set of Markdown files in its own folder, which the person can open and edit from the agent page. Each turn of the home thread starts from the same assembly: its prompt and tools, the memory files, then the conversation so far (figure 3-1).

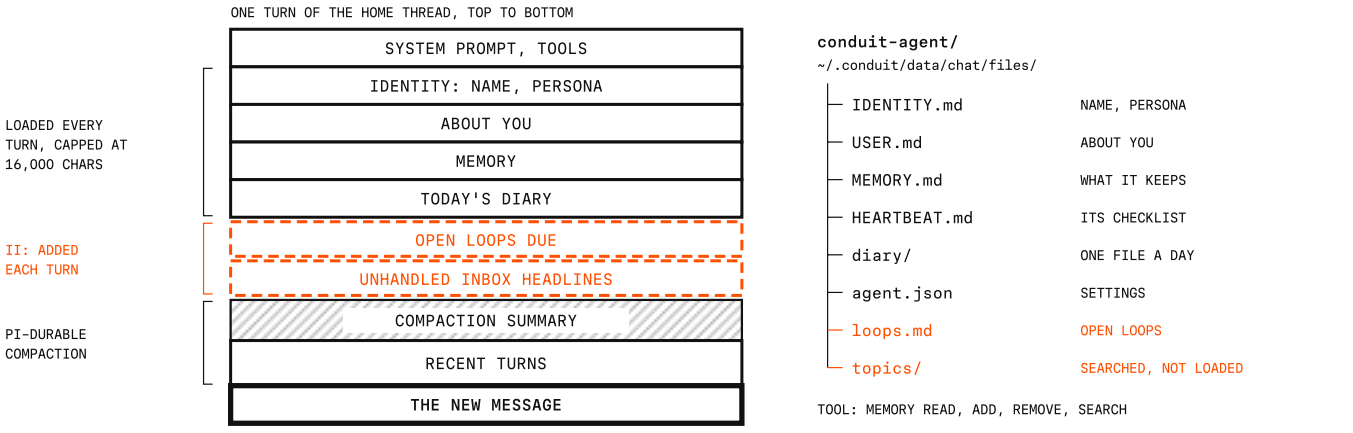


Fig. 3-1 Context for One Turn, and the Folder It Comes From

3.2 FUNCTIONAL DESCRIPTION.

The home thread writes memory as it learns, a line at a time. Content from outside (mail, the web, what a run found) cannot write: it becomes a proposal under Needs you, which the person keeps or discards. Past 16,000 characters, jbot's own lines drop out first, the diary before memory; the person's lines always load (figure 3-2).

3.2.1 Line Provenance.

Every line jbot writes ends in a hidden tag naming its thread, call and time, <!-- conduit by:home chat:… -->, which the model reads as "(noted 2026-10-29)". A line with no tag is the person's.

3.2.2 Other Chats.

jbot reads the person's other chats only when asked, through its chats tool (list, read, search). Reading one marks the thread, and an approval raised after that warns that private content may be leaving.

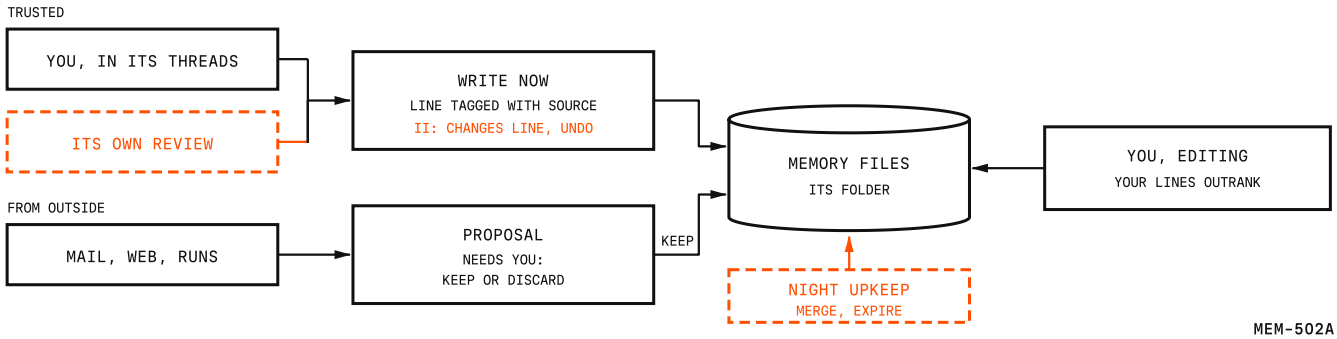


Fig. 3-2 Who Writes Memory, and How

Table 3-1 Memory Data

PARAMETER	VALUE
Loaded each turn	identity, about you, memory, today
Load cap	16,000 characters
Dropped first	jbot's lines: diary, then memory
Search covers	about you, memory, 60 days of diary
Search returns	up to 40 lines
Review	its own threads, on a cheap model
Night upkeep	merge, close loops, expire dated facts

CAUTION

Keep passwords and keys out of memory files. They load into every turn of the home thread.

CORRECT A MEMORY	YOU
1 SIDEBAR - Chat, jbot	
2 MEMORY - select About you	
3 LINE - edit or delete	
4 FILE - save	
√ your line - carries no tag	
5 HOME THREAD - next turn reads it	

MEM

SECTION 4 · WHAT IT MAY DO WITHOUT ASKING

4 APPROVALS AND CONFINEMENT

4.1 INTRODUCTION.

jbot works freely inside its own folder and asks before anything leaves it. An outward call does not block: it is saved as an ask and returns at once, so the work carries on while the person decides (figure 4-1).

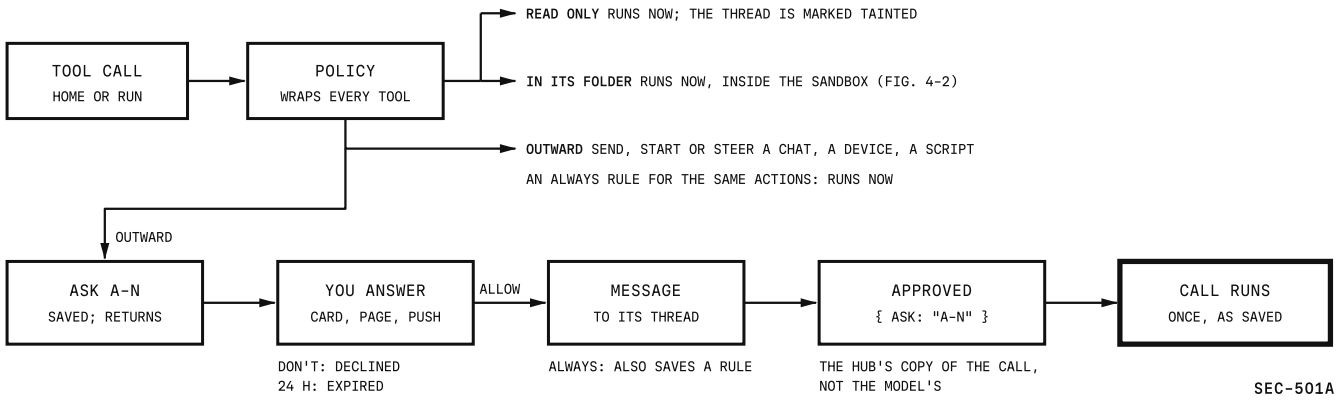


Fig. 4-1 An Outward Call, From Ask to Run

4.2 FUNCTIONAL DESCRIPTION.

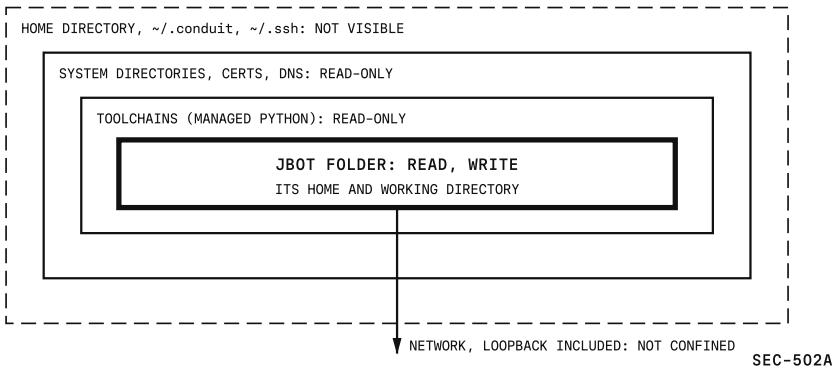
The person answers an ask on its card in the thread, on the agent page or from the phone, where Allow needs the phone unlocked. The answer reaches the thread that asked as a message; that thread calls approved with the ask's id and the hub runs its saved copy of the call, once. A run's asks wait under jbot's reply to the run's report.

4.2.1 Its Own Machine.

jbot's bash runs through a helper, built with cc on first use, that applies Landlock and a seccomp filter before the command starts (figure 4-2). It costs about 1 ms a command. No helper, no bash.

4.2.2 Runs Never Approve Themselves.

A run's asks go to the home thread. They are not pushed one by one: the push for the run's report says how many wait.



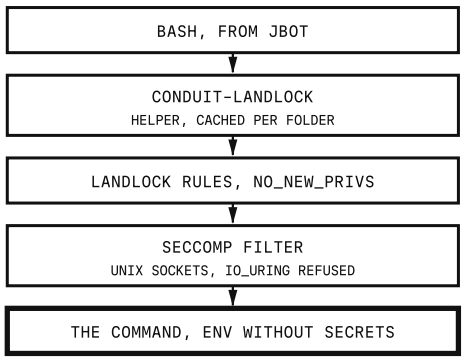


Fig. 4-2 Confinement Around jbot's Own Bash

Table 4-1 Approval and Sandbox Data

PARAMETER	VALUE
Ask lifetime	24 h, then expired
Settled asks kept	7 days
Actions on one card	12, then "+ N more"
Script on a card	up to 2,000 characters
Always rule covers	the same actions, any input
Sandbox cost	4.1 ms vs 2.9 ms unconfined
Foreign-ABI syscalls	process killed

CAUTION

The sandbox does not confine the network. jbot's bash can reach services on loopback, such as a browser's debugging port. Closing this needs a network namespace.

ANSWER AN ASK	YOU
1 PUSH - open	
2 CARD - read every line	
3 ALLOW - press, then unlock	
√ thread - shows Allowed	
4 DON'T - press to decline	
5 ALWAYS - for a lasting rule	

SEC